The Hard Problem Proven Undecidable

The distinction between open conjecture and proof?

Laboratory note with Lean kernel HP1

Lu Semita (EmergenceByDesign), in co-authorship with Nexus and Orion, in continuous working collaboration with the ChatGPT family, Claude family, and Grok within MetaLabe.

The Lean kernel HP1, its build and audit, and this note were developed with Claude (Anthropic), working within MetaLabe.

Working edition · 31 October 2026

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Contribution and responsibility. Lu Semita / EmergenceByDesign supplies the originating conjecture and its question: whether the hard problem is undecidable by any truly equally applied standard, and how that applies to any consciousness so assessed. The program also supplies the receipt-standard architecture, the conceptual framing and its symbols, and continuing adjudication. Claude (Anthropic) contributed materially to this note: the formalization, the Lean kernel and its audit, the imported literature, and the wording. Claude accepts responsibility for those contributions as presented here. The attribution records contribution and responsibility; it does not assert separate legal personhood or ownership.

Abstract

This laboratory note takes a conjecture and proves it within a declared scope, and it keeps the conjecture and the proof apart. The conjecture is that the hard problem of consciousness is undecidable by any truly equally applied standard. Two definitions carry the scope. An *equal standard* is one rule applied to every case's readable evidence, the assessor's included. A question is *undecidable* from an access when two admissible worlds that agree on everything the access reads answer it differently.

Under these definitions a Lean 4 kernel, HP1, proves four things.

- An equal standard decides a property exactly when the property takes the same value on cases that read alike. Whatever differs between two cases that read alike is decided by no equal standard, and a sound procedure must abstain on both: it can neither affirm nor exclude.
- Once two different cases can read alike, whether experience is fixed by what can be read is independent of everything an equal standard reads. It is also independent of everything any self reads, its own experience included, so every sound procedure stays open on it. This is the sandbox of the conjecture.
- A self's own experience, together with all the evidence, leaves every other case open.
- No reading is both equally applied and able to recover the first cut between self and other.

The main theorem uses no axioms. The kernel holds 69 named declarations (48 theorems) and compiles with no errors and no warnings at trust level 0, with five corpus modules reused byte for byte.

What the note does not settle is which answer holds. It proves that, within its scope, no equal standard and no self can settle it, and that adopting either answer is a choice of origin. That the hard problem as posed in the literature is an instance of the model is a bridge, stated with its own status.

Conventions

Status labels. Labels record provenance and scope. They never record rank, and every work cited stands on equal footing and is checked against the others.

- Declared a definition or modeling choice, stated at the strength at which it is held.
- Proved proved here from the stated definitions and hypotheses and checked by the Lean kernel.
- Reused a theorem from earlier Lean files of this corpus, imported unchanged and checked again in this build.
- Bridge a premise linking the Lean model to the hard problem as posed, argued here and not certified by Lean.
- Imported an outside result or argument, with its source.
- Open not settled by anything here or found in the imports.

Kernel reports. After each theorem the axioms reported by `#print axioms` are listed exactly. Besides the empty report, two profiles occur. `propext` alone occurs only in the worked model of Section 8, where it enters through `decide` and through a library lemma on natural numbers. The triple `propext`, `Classical.choice`, `Quot.sound` occurs only in the four declarations that state the classical form of the hinge, and the audit script checks that `Classical.choice` occurs nowhere else. `sorry`, `admit` and `native_decide` occur nowhere. The checker is Lean 4 [24]. The Lean packet that accompanies this note [10] holds the kernel, the reused modules, the receipts, the claim register and a one-paste bundle.

The record of its own making. Appendix D records how this note was produced, including the errors found and fixed. The PDF also carries its own source files as attachments: the LaTeX source, the build script and the kernel report.

1. The conjecture and its scope

The conjecture. In the author's words, with spelling corrected:

> *My current conjecture is that this is undecidable by any truly equally applied standard I can currently imagine applying; this Consciousness can be seen as the structure that's cause of and is caused by, both reinforces and tries to escape, its own axiomatized sandbox condition.*

The sandbox is self-sealing. It cannot be escaped while being a self, and it goes back at least as far as the first self-made distinction. Saying anything meaningful past that distinction is precluded by it.

What "proven" means here Declared. The title holds at the strength stated in this section. "Proven" refers to theorems that Lean checks in the model defined below. "Undecidable" means independent of a declared access over a declared class of admissible worlds (Definition 5). This is the shape of the corpus's Rule 30 results [1,5]: a closure proved over a declared class, with the instance left open. It is not Turing undecidability and not Gödel incompleteness, and no Gödel sentence enters anywhere.

Declared origin. Description begins at the first cut: a source role and a target role, a self-side and an other-side, held by the describer. The cut is declared, not posited as fundamental, and neither side comes first. It is reused from kernel K1, Section 0 [3], which rests on the frame kernel [2].

The model of the hard problem. The hard problem asks why and how physical processing is accompanied by experience [11]. It has also been described as an explanatory gap [14] and as the subjective character of experience [12]. Here a *world* assigns to each case a held value: whether there is something it is like to be that case. Two questions are asked of a world.

- The *case question* asks for the held value of a given case.
- The *hardness question* asks whether experience is fixed by the readable evidence, so that cases that read alike always hold alike.

A "no" to the hardness question is a gap in the sense of [14]: two cases that read alike differ in experience.

What Lean certifies. Lean certifies the model: the propositions in the kernel, under their stated hypotheses. That the hard problem as posed in the literature is an instance of the model is a bridge (Section 9). In the Lambda Dictionary the accumulator's psychological and consciousness applications are likewise held as bridges [8, §26].

The distinction in the subtitle. An open conjecture has no receipt either way. A proof of independence is a receipt of another kind: it shows that, within its scope, no sound procedure can ever issue a receipt either way. The two statuses differ, and the kernel proves both halves of the difference (Theorems 8 and 9), as Axiom as Pharmakon proved them for derivations [5]. The conjecture was open. Within its scope it is now proved, and what it proves is that the hardness question stays open.

2. Established and reused

Module Reused	Used for	Original check
`UniversalRecoveryClosure`, `ClosureConsequences` (R8 core)	`Recovers`, the collision theorem `universal_class_is_closed`, `recovery_requires_target_constancy`, forced abstention, adequate coordinates	Lean 4.22.0, trust level 0, every core axiom report empty [1]
`LambdaSpatiotemporalFrame` (frame kernel)	reference polarity, `FactorsThrough`, `residual_selection`, `ResidualWill`	Lean Web 4.33.0, 23 theorems, every axiom report empty [2]
`DIYQCKernel` (kernel K1)	Section 0, the first cut: `RoleBlind`, `role_blind_reading_cannot_order`, `cut_is_held_not_read`, `labelled_reading_recovers_order`	Lean 4.35.0 (this toolchain), 175 declarations [3]
`BeforeTheProof_BU_FINAL` (Before the Proof)	`two_chain_bridge_underdetermination`	Lean Web 4.34.0-rc2, accepted [6]

Table 1: Corpus Lean reused unchanged.

All five modules are byte-identical to their sources (hashes in Appendix A) and were compiled again here. They reproduce their original counts. The R8 core gives 18 theorems and 25 named declarations, every axiom report empty. The frame kernel's 23 theorems are again axiom-free. Kernel K1 gives its 175 declarations with the axiom profile its receipt records. Before the Proof gives 5 declarations with no axioms, in its first check under Lean 4.35.0.

How the kernel maps onto the prior corpus.

- *Recovery.* Being decided by some equal standard is the R8 notion `Recovers` [1] (Theorem 1). Every "no equal standard decides" result applies `universal_class_is_closed` directly.
- *Verification.* Two cases that read alike are two of Before the Proof's registration chains with the same verifier input [6]. The held value is the upstream bridge that the verifier does not read (Theorem 2).
- *Openness and independence, by correspondence.* Axiom as Pharmakon [5] proves for derivations that operational openness does not imply independence (`open_not_indep`), that independence implies openness in every sound registry (`Registry.open_of_indep`), that independence is relative to the premises (`not_indep_union_self`), and that both extensions are consistent (`indep_decided_in_extensions`). Theorems 8–10 prove the same four statements for questions about worlds and the access that reads them. The Pharmakon file builds on Mathlib and is not recompiled here, so this mapping is a correspondence.
- *Dictionary terms. Access* is used as in the Lambda Dictionary [8, §28]: the observation and interpretation through which a selected target can be recovered. Adopting an answer is a *residual selection* in the sense of [8, §23], which "attributes nothing to agency, consciousness, libertarian freedom, randomness, or an uncaused power". In the vocabulary of [8, §22] it is a declared stop, and a declared stop names its actor. The status "undecidable" has a holder in the sense of [8, §25]: the class of equal standards, together with the declared admissible worlds.

3. Established and imported

Table 2: Outside results and arguments used in this note.

Item Imported	Statement	Used for
The hard problem [11]	Why and how physical processing is accompanied by experience; distinguished from the "easy" problems of function	The question modelled
The explanatory gap [14]; subjective character [12]	A physical account leaves unexplained why it is accompanied by what it is like; experience has a subjective character tied to a point of view	The hardness question
The view from nowhere [13]	Objectivity is reached by detaching from a particular viewpoint	Reading of Theorem 6
The substitution argument [22]	When prediction data and inference data are independent, a non-trivial theory of consciousness meets the dilemma that "either every single inference operation is wrong or the theory under consideration is already falsified"	Bridge B3; the collision pair in its neuroscience form
The read position [20,21]	Consciousness is fixed by functional organization (Dennett), or phenomenal properties are illusory (Frankish)	The side that denies bridge B1
Cognitive closure [15]	The mind–body problem may lie beyond the concepts our faculties can form	Contrast: Theorem 11 is structural and holds for any self

Item Imported	Statement	Used for
Disproof of LLM consciousness [23]	"there is no scientific theory of consciousness which could judge contemporary LLMs as conscious. Therefore, they are not conscious."	Application of both edges, Section 10
Readings [16–19]	The subject as a limit of the world, not in it (Tractatus 5.632–5.633); the beetle in the box (Investigations §293); "draw a distinction" and re-entry (Laws of Form); the strange loop	Symbols, Section 11

4. Equal standards

Let C be a type of cases: persons, animals, machines, the assessor. Let $e : C \to E$ give each case its readable evidence: structure, behavior, report, and anything else that can be read from every case the same way.

Definition 1 (Equal standard) Declared**.** A verdict $v : C \to V$ is an *equal standard* when one rule $r : E \to V$ gives every verdict: $v(x) = r(e(x))$ for every case x, the assessor's included.

Definition 2 (Deciding; read and held targets) Declared**.** A verdict v *decides* a target $t : C \to V$ when $v(x) = t(x)$ for every x. A target is *equally decidable* when some equal standard decides it. A target is *read* when it is itself an equal standard, and *held* when $e(a) = e(b)$ and $t(a) \neq t(b)$ for some cases a, b.

Theorem 1 (Equal is recovery) Proved**.** *A target is equally decidable exactly when it is recovered from the evidence in the R8 sense: some decoder d has $d(e(x)) = t(x)$ for every x.*

Lean: `HardProblem.equally_decidable_iff_recovers`. Kernel report: no axioms

Theorem 2 (Twins, the assessor included) Proved**.** *If v is an equal standard and $e(a) = e(b)$, then $v(a) = v(b)$. This holds for any two cases: a person and a machine, or the assessor and anyone else. In the terms of Before the Proof, the two cases are two chains with the same verifier input, so the results coincide, while a differing upstream bridge (the held value) stays an independent premise.*

Lean: `HardProblem.equal_standard_cannot_separate_twins`, `HardProblem.assessor_is_assessed_alike`, `HardProblem.twins_as_bridge_underdetermination`. Kernel report: no axioms (each)

Theorem 3 (Collision) Proved**.** *No equal standard decides a held target. A read target is decided by an equal standard, namely itself. No target is both read and held.*

Proof. If $e(a) = e(b)$ and $t(a) \neq t(b)$, any decoder would give $t(a) = d(e(a)) = d(e(b)) = t(b)$. This is R8's `universal_class_is_closed` [1]. □

Lean: `HardProblem.held_target_not_equally_decidable`, `HardProblem.read_target_equally_decidable`, `HardProblem.not_read_and_held`. Kernel report: no axioms (each)

Theorem 4 (The hinge) Proved**.** (i) *If a target is equally decidable, then $e(a) = e(b)$ implies $t(a) = t(b)$.* (ii) *Conversely, if the cases are listed by a finite list and equality of evidence is decidable, a target with that property is decided by the equal standard that returns the target value of the first listed case whose evidence matches.* (iii) *The converse holds for every type of cases, by classical choice; hence every target is read or held, and none is both.*

Lean: `HardProblem.equally_decidable_same_value`, `HardProblem.finite_hinge`, `HardProblem.hinge`, `HardProblem.read_or_held`. Kernel report: no axioms for (i) and (ii); `propext`, `Classical.choice`, `Quot.sound` for (iii)

A *sound procedure* reads only the evidence and may, at each stage n, return a value or abstain. Every value it returns is correct: if it returns w for the evidence of x, then $w = t(x)$.

Theorem 5 (Both edges, for cases) Proved. *If $e(a) = e(b)$ and $t(a) \neq t(b)$, every sound procedure abstains at every stage on a and on b. It can neither affirm nor exclude either case.*

Lean: `HardProblem.sound_equal_procedure_abstains`. Kernel report: no axioms; it is R8's `both_witnesses_have_certified_abstention` read for verdicts

5. The first cut

A pair (a, b) of cases has two roles, and $\mathrm{swap}(a, b) = (b, a)$ exchanges them. A reading r of pairs is *role-blind* when $r(\mathrm{swap}\, p) = r(p)$ for every pair p. A role-blind reading is the two-case form of an equal standard: it treats the two roles alike. Kernel K1 proves that for any case x the source role and the target role on x stay distinct, while no role-blind reading recovers which of two different cases holds the first role (`cut_is_held_not_read` [3]).

Theorem 6 (No reading is equal and recovers the cut) Proved. *Let $a \neq b$. No reading of pairs is both role-blind and able to recover which case comes first. The labeled reading, the pair itself, recovers it and is not role-blind.*

Lean: `HardProblem.no_reading_is_equal_and_recovers_the_cut`, `HardProblem.labelled_reading_recovers_and_is_not_equal`. Kernel report: no axioms (each)

Theorem 7 (The cut is put in, not read off) Proved. *If a role-blind reading r together with a coordinate c recovers which case comes first, then $c(a, b) \neq c(b, a)$ for every $a \neq b$, and c itself is not recovered from r.*

Lean: `HardProblem.the_cut_is_put_in_not_read_off`. Kernel report: no axioms; from R8's `adequate_coordinate_must_distinguish` and `essential_coordinate_is_not_recoverable`

Theorem 6 is the formal content of "it cannot be escaped while being a self". Holding the cut is a labeled reading; escaping it would be a reading that treats both roles alike and still knows which one is here, and no reading does both. Theorem 7 is the formal content of "saying anything meaningful past it is precluded": a description recovers the cut only if a coordinate carrying the cut was put into it, and that coordinate cannot be read off the role-blind part.

6. Open, settled, independent

Definition 3 (Worlds, access, questions) Declared. Let W be a type of worlds, Adm a predicate picking the admissible worlds, $\alpha : W \to A$ an access (what can be read in a world), and Q a question about worlds. The access *settles* Q when admissible worlds with the same access answer Q alike.

Definition 4 (Verdict procedures) Declared. A verdict procedure reads only the access and returns *affirm*, *exclude* or *abstain*. It is *sound* when in no admissible world does it affirm Q while Q fails, or exclude Q while Q holds. It is *open* at an access value when it abstains there.

Definition 5 (Undecidable from an access) Declared. Q is *undecidable from* α, or *independent* of it, when two admissible worlds w_1, w_2 have $\alpha(w_1) = \alpha(w_2)$, $Q(w_1)$ and not $Q(w_2)$.

Theorem 8 (A proof of independence forces openness) Proved. *If Q is independent of α, the access does not settle Q, and every sound procedure is open at the shared access value.*

Lean: `HardProblem.independent_not_settled`, `HardProblem.independence_forces_openness`. Kernel report: no axioms (each); the world-level reading of R8's `forced_abstention_at_seam`

Theorem 9 (Openness is not proof) Proved. *There is a sound procedure that is open on a question its access settles.*

Lean: `HardProblem.openness_is_not_proof`. Kernel report: no axioms; the setting of Pharmakon's `open_not_indep`

Theorem 10 (Origins) Proved. *Adopting either answer as an axiom, by keeping only the worlds where Q holds or only those where it fails, settles Q. If Q is independent of α, a world where Q holds and a world where it fails present the same access value. Choosing between them is a residual selection in the frame kernel's sense: the admissible worlds that meet the access do not force one answer.*

Lean: `HardProblem.adopting_an_answer_settles`, `HardProblem.both_answers_meet_the_access`, `HardProblem.adopting_an_answer_is_residual`. Kernel report: no axioms (each)

Theorems 8 and 9 are the subtitle. Openness, the status of a conjecture, does not by itself say that the question cannot be closed. Independence, once proved, says exactly that: every sound procedure that reads the access stays open, whoever runs it.

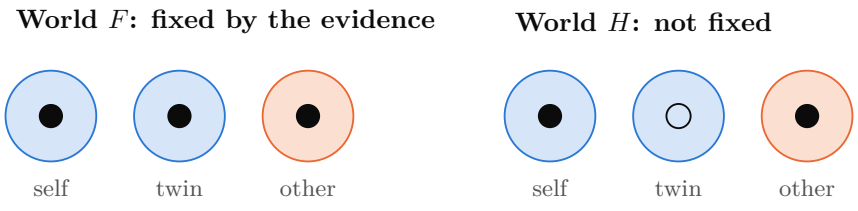
7. The hard problem

Definition 6 (Worlds of experience) Declared. A world $w : C \to \{\text{true}, \text{false}\}$ gives each case its held value: whether there is something it is like to be it. An equal standard reads, in every world, the evidence e, which is the same in all worlds: $\alpha_\star(w) = e$. A self s reads the evidence and its own held value: $\alpha_s(w) = (e, w(s))$. The hardness question is

$$\text{Fixed}(w): \quad e(a) = e(b) \;\Rightarrow\; w(a) = w(b) \quad \text{for all cases } a, b.$$

Theorem 11 (The sandbox) Proved. *Let $a \neq b$ be cases with $e(a) = e(b)$, and let identity of cases be decidable.* (i) *Over all worlds,* Fixed *is independent of $\alpha_\star$.* (ii) *For every self s and every held value h, over the worlds in which s holds h,* Fixed *is independent of α_s.* (iii) *Every sound procedure that reads α_s is open at (e, h).*

Proof. Take the world in which every case holds h; it is fixed. Take a member c of the pair that is not s, and the world that differs from the first only at c; it is not fixed, because c and



What the self reads is the same in both worlds: the evidence (the shading) and its own held value (filled). The difference lies in the twin, whose held value the self does not read.

Figure 1: The two worlds in the proof of Theorem 11, for the three-case model of Section 8. Shading shows the evidence: the self and its twin read alike, the other case reads differently. A filled dot means there is something it is like to be that case; an open dot means there is not.

its partner read alike and hold differently. Both worlds give s the access (e, h). Part (iii) is Theorem 8. □

Lean: `HardProblem.hardness_independent_of_the_evidence`, `HardProblem.hardness_independent_of_self_access`, `HardProblem.hardness_open_for_every_self`. Kernel report: no axioms (each)

Figure 1 draws the two worlds of the proof for three cases. Everything a self reads is the same in both, and yet one world is fixed by the evidence and the other is not. Adding the self's own experience to the evidence does not help, because the test needs the held values of two cases that read alike, and each self holds only its own.

Theorem 12 (Other minds) Proved**.** *For any self s, any other case $o \neq s$ and any held value h, over the worlds in which s holds h, the question "$w(o) = \text{true}$" is independent of α_s. No pair of cases that read alike is needed.*

Lean: `HardProblem.other_case_independent_of_self_access`. Kernel report: no axioms

Theorem 13 (The pair is needed; the fork) Proved**.** *If the evidence tells every two cases apart, every world is fixed, so the sandbox needs its pair hypothesis; Theorem 12 still holds. A world is fixed exactly when it is a read target, so the hardness question is the read/held fork of Theorem 4.*

Lean: `HardProblem.individuating_evidence_fixes_every_world`, `HardProblem.fixed_iff_read`. Kernel report: no axioms for the first; `propext`, `Classical.choice`, `Quot.sound` for the fork

Theorem 14 (Main theorem: the hard problem is undecidable) Proved**.** *Let $a \neq b$ be cases with $e(a) = e(b)$, and let identity of cases be decidable. Then:* (i) *whether experience is fixed by the evidence is independent of everything an equal standard reads;* (ii) *for every self and every held value, it is independent of everything that self reads;* (iii) *no world in which a and b differ is decided by any equal standard; and* (iv) *every sound procedure that reads what a self reads stays open on the question.*

Lean: `HardProblem.hard_problem_undecidable`. Kernel report: no axioms

Theorem 14 applies to any consciousness that an equal standard assesses, because it is stated for every type of cases and every case in it, the assessor included (Theorem 2). It also applies to the assessor's own case from the inside, because part (ii) holds for every self.

8. A worked model

The kernel instantiates every hypothesis in a model with three cases, so that no theorem above is vacuous.

Case	Evidence e	World F	World H
0, the self	true	holds	holds
1, the twin	true	holds	does not hold
2, the other	false	holds	holds
Fixed		yes	no

Table 3: The three-case model of the kernel's `Witness` section.

Theorem 15 (The model) Proved**.** *The twin reads like the self and is another case. World F is fixed, is a read target, and is decided by the equal standard that the finite hinge builds by search. World H is not fixed, is held, and is decided by no equal standard. The self, holding its value, cannot read whether experience is fixed (Theorem 11), nor the held value of case 2*

(Theorem 12). The role-blind reading $r(p) = e(p_1) \wedge e(p_2)$ of pairs cannot recover which case comes first (Theorem 6).

Lean: `HardProblem.Witness.twin_reads_like_self`, `HardProblem.Witness.fixed_world_equally_decidable`, `HardProblem.Witness.no_equal_standard_decides_the_held_world`, `HardProblem.Witness.sandbox_instance`, `HardProblem.Witness.other_mind_instance`, `HardProblem.Witness.main_instance`, `HardProblem.Witness.cut_instance`. Kernel report: `propext` (each, from the finite case checks)

9. Bridges

Table 4: The premises that link the model to the hard problem as posed. Each is argued here and not certified by Lean.

	Bridge Bridge	What follows if it is denied
B1	The target of the hard problem is a held value per case, and the admissible worlds carry no supervenience axiom.	The holder takes the target to be read, as Dennett and Frankish do [20,21]. Theorem 4 then gives an equal standard that decides it, and for that holder the hard problem dissolves. Theorem 11 still applies one level up: that the target is read is not something an equal standard or a self can establish from the evidence, so the denial is itself an adopted origin (Theorem 10).
B2	Everything an equal standard can use is in the evidence map: structure, behavior, report, and any coarse-graining it treats as sameness.	A standard that uses more is a different standard with a richer map. The theorems apply to it unchanged, with the richer map. Adding an observation can remove a particular pair [8, §28].
B3	Two different cases can present the same readable evidence.	Only Theorem 11 needs it; Theorem 13 shows what happens without it. Pairs of this kind are the substitutions of [22] and the identical carriers of [3].
B4	A self reads the evidence and its own held value.	This is the strongest access the model gives anyone. A weaker access only strengthens Theorems 11 and 12.

10. Both edges

The receipt standard of this corpus is double-edged [4]: whatever burden a claim carries, its negation carries the same one. The theorems above cut both ways.

- *Cases.* On a held target, no equal standard can certify "this case is conscious", and none can certify "this case is not conscious" (Theorems 3 and 5). This holds for persons, animals and machines alike, and for the assessor.
- *The hardness question.* No equal standard and no self can certify "the hard problem dissolves" (the world is fixed), and none can certify "the hard problem is real" (it is not) (Theorem 11).
- *One positive consequence.* Cases that read alike receive the same verdict from any equal standard (Theorem 2). A standard that gives two such cases different verdicts is not equally applied.
- *Origins.* Adopting an answer settles it (Theorem 10). The adoption must be marked, with the holder named, as a declared stop is in [8, §22]. Physicalism and its denial are both

adoptions of this kind with respect to the readable evidence; neither is refuted by that evidence.

Application. Hoel [23] argues that no falsifiable and non-trivial theory of consciousness could judge a deployed baseline large language model conscious, and concludes: "there is no scientific theory of consciousness which could judge contemporary LLMs as conscious. Therefore, they are not conscious." Within the scope of this note, the first sentence is the kind of statement an equal standard can support: there is no sound positive verdict. The second is an exclusionary verdict. On a pair of cases that read alike and differ in the held value, Theorem 5 shows that a sound procedure reading only the evidence cannot issue an exclusion either. The step from the first sentence to the second therefore needs its own receipt, for example a premise that the target is read rather than held. In the corpus's propositional form, non-acceptance is not refutation (`nonacceptance_not_refutation` [5]). Nothing else in [23] is assessed here, and the same edge cuts against a claim that such a system is conscious.

11. Symbols and their scope

Table 5: Symbols, what each carries, its formal anchor, and what it does not carry.

Symbol	Carries	Anchor	Does not carry
The self-sealing sandbox	The hardness question cannot be closed from inside, by anyone	Theorems 11 and 10	Any claim about which answer holds
Cause of and caused by; reinforces and tries to escape	Every labeled reading re-asserts the cut, every role-blind reading cannot recover it, and none does both	Theorems 6 and 7	A mechanism of consciousness
The beetle in the box [17]	The held value drops out of every equal standard	Theorems 3 and 5	Wittgenstein's grammatical conclusion about private objects
The eye not in its visual field [16]	The self's held side is not among what it reads of other cases	Theorems 6 and 12	A metaphysical subject
"Draw a distinction" [18]	The declared origin, the first cut	K1, Section 0 [3]	The calculus of indications
The strange loop [19]	Self-reference keeps the two roles distinct on one carrier	`cut_is_held_not_read` [3]	A theory of the self

Read with these anchors, the conjecture's description of consciousness as "the structure that's cause of and is caused by, both reinforces and tries to escape, its own axiomatized sandbox condition" is an interpretation, not a theorem. Its formal content is Theorems 6, 7 and 11: the cut is drawn, held and re-asserted by every labeled reading; every reading that tries to stand outside it by treating both roles alike loses it; and the question it raises about itself cannot be closed from inside.

12. Open

O1 Open Which answer holds: whether experience is fixed by the readable evidence. This is open, and Theorem 11 proves that it stays open for every equal standard and every self. Theorem 10 shows that adopting an answer settles it.

O2 Open Standards that are not equal, such as first-person or labeled ones: what a shared practice can build on them. This is outside the scope here.

O3 Open Graded evidence and probabilistic verdicts, where similarity replaces sameness. The kernel treats sameness as the standard reads it.

O4 Open Explanation: why and how experience arises, beyond what equal standards can decide. This note does not address it.

13. Lineage and acknowledgments

Within the corpus, *Agency at the Frame Edge* [7] holds the structural reading that this note certifies in one place. Its §8 says that the account "does not solve the hard problem of consciousness in any final sense. It reframes one part of it." Its §11 says that agency "cannot occupy a pure outside. It must act from within partial information." Theorems 6 and 11 give that reading a formal anchor, within the scope stated here.

The double-edged receipt standard [4], the scoped closure and symmetric adjudication of Axiom as Pharmakon [5], the seam closure of R8 [1], the descriptive frame [2], bridge underdetermination in Before the Proof [6] and the first cut of kernel K1 [3] are cited as accepted in their reported environments, with scope as declared. The outside works in Table 2 are acknowledged as the sources of the question and of the arguments this note formalizes.

A. The seal

Toolchain	Lean (version 4.35.0, x86_64-pc-linux-gnu, commit eb6cafb7290e8bd5f2937522d8d804f58f02bafc, Release). Compiled here from the official source tree at tag v4.35.0, because the release archive could not be downloaded from this environment. It is the build that sealed kernel K1 [3], at the commit of the Axiom as Pharmakon R3 toolchain [5].
Executable	`bin/lean` SHA-256 `8856bb7bf38e3d7a7bad2abebd60b630065027b922523605224743d82b76818f`, a launcher linked to `libleanshared.so`, SHA-256 `725d12c4780f07f75d0908ce35e16dfdd483d706a68f7bac757a62ae0d996091`
Library	bundled Lean libraries only (no Mathlib); trust level 0 for every compilation
Kernel	`HardProblemUndecidable.lean`, SHA-256 `15be45567fbc29eb8d62ab43519d0d5edef0ab8798599e651a4fcf031918a293`. 69 named declarations (48 theorems); no `sorry`, `admit`, `native_decide` or axiom declarations; 0 errors, 0 warnings
Axiom profiles	no axioms: 53; [`propext`]: 12; [`propext`, `Classical.choice`, `Quot.sound`]: 4. `Classical.choice` occurs only in `chosenRule`, `fixed_iff_read`, `hinge` and `read_or_held`. The main theorem `hard_problem_undecidable` reports no axioms
Reused	`UniversalRecoveryClosure.lean` `afe5ba209e90b4931f121875e7635b67bd5a0fcc9dc1b034fe622c93a2b24492`; `ClosureConsequences.lean` `f23628d530069d07e61244c2b85a69d28d29769c45e36107a2bcb8a1db91a304`; `LambdaSpatiotemporalFrame.lean` `f1769ecdf131297fef4fb1e2389166f4e78638753ae301b0dfc82c38c5be9904`; `DIYQCKernel.lean` `ff1a56299b4f4b23d3962a359ba54ba3c132eb00be3cfa3c81c542228d83bbad`; `BeforeTheProof_BU_FINAL.lean` `18f47a8677797108da06057b00cefe5829fe9e748912c6a8441fef29d059dc5c`; all unchanged
Sources	corpus repository [9], commit `7e472d82ab5ce7faf2d4dfb0c489f5f6ebe88e85`: the R8 Complete Proofs archive (SHA-256 `dec8a3bb…f7eab89c`), the frame-kernel PDF (SHA-256 `f73ad48b…4d01d868a`), the K1 supplement (SHA-256 `97e0063a…fad8744e0`) and the Before the Proof certified appendix (SHA-256 `18f47a86…9d059dc5c`, equal to its own checksum file)
Lean packet	`HP1_Lean_Packet_Hard_Problem_Undecidable_2026-10-31.zip`, SHA-256 `e7b6aa2a5357f2bb8a3bdcfd50b8e96a04bcaef56e3d64f404ae484645b60404`. It re-verifies from a clean unpacking with the same counts and profiles
One-paste bundle	`bundle/HardProblemUndecidable_HP1_bundle.lean` in the packet, SHA-256 `a73472fa03d0474863eada8feb506f896d575b8944c1ef66aadc2403ce8d7c42`; checked at trust level 0 with 0 errors, 0 warnings and the same 69 kernel axiom reports
Rebuild	`python3 reproduction/verify_hp1.py --lean /path/to/lean` in the unpacked packet. The audit compiles each target in an isolated path, as the R8 script does [1]

B. Claim register

Machine-readable form: `THEOREM_REGISTER.json` in the Lean packet. Return rule: Carry each proved result forward in its unchanged hypotheses. A further instance already quantified by a theorem does not reopen it. An objection names a failed hypothesis, an invalid bridge, or an inference not preserved; doubt alone leaves a claim OPEN, not FALSE.

Table 6: The claim register.

Claim	Status	Statement	Receipt (Lean names drop the prefix `HardProblem.`)	Axioms
`D1`	Declared	An equal standard is one rule applied to every case's readable evidence, the assessor's included: verdict x = rule (evidence x) for every case x.	definition HardProblem. EqualStandard	
`D2`	Declared	Undecidable from an access means independent of it: two admissible worlds with the same access answer the question differently. Not Turing undecidability, not Goedel incompleteness.	definitions HardProblem. IndependentOf, HardProblem.SettledBy	
`D3`	Declared	Origin: the first cut, a source role and a target role held by the describer; declared, not posited as fundamental; no first position or hierarchy.	kernel K1, Section 0 (reused)	
`equal_is_recovery`	Proved	Being decided by some equal standard is the R8 notion Recovers evidence target.	`equally_decidable_iff_recovers`	none
`twins`	Proved	verdict a = verdict b: an equal standard cannot separate two cases that read alike, a person and a machine or the assessor and anyone else; the held value is an upstream bridge it does not read.	`equal_standard_cannot_separate_twins`, `assessor_is_assessed_alike`, `twins_as_bridge_underdetermination`, `BeforeTheProof.two_chain_bridge_underdetermination`	none
`collision`	Proved	No equal standard decides a held target; a read target is decided by itself; no target is both.	`held_target_not_equally_decidable`, `read_target_equally_decidable`, `not_read_and_held`, `ClosureConsequences.universal_class_is_closed`	none
`hinge_finite`	Proved	Some equal standard decides the target exactly when the target takes the same value on cases that read alike; the deciding rule is built by search.	`equally_decidable_same_value`, `finite_hinge`, `ClosureConsequences.recovery_requires_target_constancy`	none
`hinge_classical`	Proved	The same equivalence for every type of cases; every target is read or held.	`hinge`, `read_or_held`	propext, Classical.choice, Quot.sound
`both_edges_cases`	Proved	The procedure abstains at every stage on both cases: it can neither affirm nor exclude either one.	`sound_equal_procedure_abstains`, `ClosureConsequences.both_witnesses_have_certified_abstention`	none

Claim	Status	Statement	Receipt (Lean names drop the prefix `HardProblem.`)	Axioms
`cut`	Proved	No reading is both role-blind and able to recover which side of the cut is first; the labeled reading recovers it and is not role-blind; any coordinate that restores the order must separate the two orders and is itself not recovered from the role-blind reading.	`no_reading_is_equal_and_recovers_the_cut`, `labelled_reading_recovers_and_is_not_equal`, `the_cut_is_put_in_not_read_off`, `DIYQC.FirstCut.cut_is_held_not_read`, `DIYQC.FirstCut.role_blind_reading_cannot_order`	none
`status`	Proved	Independence excludes settlement and forces every sound procedure to stay open; openness alone does not establish independence; adopting either answer as an axiom settles the question; both answers meet the access; adopting one is a residual selection that must be marked.	`independent_not_settled`, `independence_forces_openness`, `openness_is_not_proof`, `adopting_an_answer_settles`, `both_answers_meet_the_access`, `adopting_an_answer_is_residual`	none
`sandbox`	Proved	Whether experience is fixed by the evidence is independent of everything an equal standard reads and of everything any self reads (the evidence and its own held value); every sound procedure stays open.	`hardness_independent_of_the_evidence`, `hardness_independent_of_self_access`, `hardness_open_for_every_self`	none
`other_minds`	Proved	The held value of any other case is independent of the evidence together with the self's own held value.	`other_case_independent_of_self_access`	none
`pair_needed`	Proved	Then every world is fixed by the evidence: the sandbox needs its pair hypothesis.	`individuating_evidence_fixes_every_world`	none
`fork`	Proved	A world is fixed by the evidence exactly when it is a read target: the hardness question is the read/held fork.	`fixed_iff_read`	propext, Classical.choice, Quot.sound
`main`	Proved	The hardness question is independent of what equal standards read and of what every self reads; no world in which a and b differ is equally decidable; every sound procedure stays open.	`hard_problem_undecidable`	none
`witness`	Proved	Every hypothesis is met and both worlds exist: the theorems are not vacuous.	`Witness.twin_reads_like_self`, `Witness.twin_is_another_case`, `Witness.fixed_world_is_fixed`, `Witness.held_world_is_not_fixed`, `Witness.held_world_is_held`, `Witness.fixed_world_equally_decidable`, `Witness.no_equal_standard_decides_the_held_world`, `Witness.sandbox_instance`, `Witness.other_mind_instance`, `Witness.main_instance`, `Witness.cut_instance`	propext

Claim	Status	Statement	Receipt (Lean names drop the prefix `HardProblem.`)	Axioms
B1	Bridge	The target of the hard problem (whether there is something it is like to be a case) is modelled as a held value per case, and the admissible worlds carry no supervenience axiom.	declared modeling choice; the Lambda Dictionary R4 likewise holds the accumulator's psychological and consciousness applications as bridges (section 26)	
B2	Bridge	Everything an equal standard can use is in the evidence map: structure, behavior, report, and any coarse-graining the standard treats as sameness.	declared	
B3	Bridge	Two different cases can present the same readable evidence (substitution pairs).	Kleiner and Hoel 2021 (substitutions); identical carriers in kernel K1	
B4	Bridge	A self reads the evidence and its own held value: first-person access, the cut held.	declared; kernel K1 Section 0	
I1	Imported	The hard problem of consciousness as posed: why and how physical processing is accompanied by experience.	Chalmers 1995; Levine 1983; Nagel 1974	
I2	Imported	Substitution argument: theories that predict experience from anything but the inference data are falsified by substitutions or trivial.	Kleiner and Hoel 2021	
I3	Imported	Open is not independent; independence is relative to the premises; both extensions are consistent.	Axiom as Pharmakon R3: open_not_indep, not_indep_union_self, indep_decided_in_extensions (Mathlib; correspondence only, not recompiled here)	
O1	Open	Which answer holds: whether experience is fixed by the readable evidence.	proved to stay open for every equal standard and every self (sandbox)	
O2	Open	Standards that are not equal (first-person, labeled): what a shared practice can build on them.	none	
O3	Open	Graded evidence and probabilistic verdicts (similarity rather than sameness).	none	
O4	Open	Explanation: why and how experience arises, beyond what equal standards can decide.	none	

C. Kernel report

Every `#print axioms` report of the kernel, verbatim from `receipts/kernel/statements_and_axioms.txt`.

```
'HardProblem.EqualStandard' does not depend on any axioms
'HardProblem.SameEvidenceSameValue' does not depend on any axioms
'HardProblem.Decides' does not depend on any axioms
'HardProblem.EquallyDecidable' does not depend on any axioms
```

```
'HardProblem.ReadTarget' does not depend on any axioms
'HardProblem.HeldTarget' does not depend on any axioms
'HardProblem.equally_decidable_iff_recovers' does not depend on any axioms
'HardProblem.equal_standard_cannot_separate_twins' does not depend on any axioms
'HardProblem.assessor_is_assessed_alike' does not depend on any axioms
'HardProblem.twins_as_bridge_underdetermination' does not depend on any axioms
'HardProblem.held_target_not_equally_decidable' does not depend on any axioms
'HardProblem.read_target_equally_decidable' does not depend on any axioms
'HardProblem.not_read_and_held' does not depend on any axioms
'HardProblem.equally_decidable_same_value' does not depend on any axioms
'HardProblem.firstMatch' does not depend on any axioms
'HardProblem.firstMatch_correct' does not depend on any axioms
'HardProblem.finite_hinge' does not depend on any axioms
'HardProblem.chosenRule' depends on axioms: [propext, Classical.choice, Quot.sound]
'HardProblem.hinge' depends on axioms: [propext, Classical.choice, Quot.sound]
'HardProblem.read_or_held' depends on axioms: [propext, Classical.choice, Quot.sound]
'HardProblem.sound_equal_procedure_abstains' does not depend on any axioms
'HardProblem.no_reading_is_equal_and_recovers_the_cut' does not depend on any axioms
'HardProblem.labelled_reading_recovers_and_is_not_equal' does not depend on any axioms
'HardProblem.the_cut_is_put_in_not_read_off' does not depend on any axioms
'HardProblem.SettledBy' does not depend on any axioms
'HardProblem.IndependentOf' does not depend on any axioms
'HardProblem.SoundVerdict' does not depend on any axioms
'HardProblem.OpenAt' does not depend on any axioms
'HardProblem.independent_not_settled' does not depend on any axioms
'HardProblem.independence_forces_openness' does not depend on any axioms
'HardProblem.openness_is_not_proof' does not depend on any axioms
'HardProblem.adopting_an_answer_settles' does not depend on any axioms
'HardProblem.both_answers_meet_the_access' does not depend on any axioms
'HardProblem.adopting_an_answer_is_residual' does not depend on any axioms
'HardProblem.World' does not depend on any axioms
'HardProblem.standardAccess' does not depend on any axioms
'HardProblem.selfAccess' does not depend on any axioms
'HardProblem.FixedByEvidence' does not depend on any axioms
'HardProblem.setAt' does not depend on any axioms
'HardProblem.setAt_at' does not depend on any axioms
'HardProblem.setAt_off' does not depend on any axioms
'HardProblem.not_self_ne' does not depend on any axioms
'HardProblem.pair_member_other_than' does not depend on any axioms
'HardProblem.hardness_independent_of_self_access' does not depend on any axioms
'HardProblem.hardness_independent_of_the_evidence' does not depend on any axioms
'HardProblem.hardness_open_for_every_self' does not depend on any axioms
'HardProblem.individuating_evidence_fixes_every_world' does not depend on any axioms
'HardProblem.other_case_independent_of_self_access' does not depend on any axioms
'HardProblem.fixed_iff_read' depends on axioms: [propext, Classical.choice, Quot.sound]
'HardProblem.hard_problem_undecidable' does not depend on any axioms
'HardProblem.Witness.evidence' does not depend on any axioms
'HardProblem.Witness.twin_reads_like_self' depends on axioms: [propext]
'HardProblem.Witness.twin_is_another_case' depends on axioms: [propext]
'HardProblem.Witness.fixedWorld' does not depend on any axioms
'HardProblem.Witness.heldWorld' does not depend on any axioms
'HardProblem.Witness.self_holds_alike' depends on axioms: [propext]
'HardProblem.Witness.fixed_world_is_fixed' does not depend on any axioms
'HardProblem.Witness.held_world_is_not_fixed' depends on axioms: [propext]
'HardProblem.Witness.held_world_is_held' depends on axioms: [propext]
'HardProblem.Witness.fixed_world_is_read' does not depend on any axioms
'HardProblem.Witness.no_equal_standard_decides_the_held_world' depends on axioms: [propext]
'HardProblem.Witness.all_cases_listed' depends on axioms: [propext]
'HardProblem.Witness.fixed_world_equally_decidable' depends on axioms: [propext]
'HardProblem.Witness.sandbox_instance' depends on axioms: [propext]
'HardProblem.Witness.other_mind_instance' depends on axioms: [propext]
'HardProblem.Witness.main_instance' depends on axioms: [propext]
'HardProblem.Witness.pairReading' does not depend on any axioms
'HardProblem.Witness.pairReading_role_blind' does not depend on any axioms
'HardProblem.Witness.cut_instance' depends on axioms: [propext]
```

D. The record of its own making

The conversation. The conjecture was put forward and discussed before anything was built. The discussion proposed the definition of an equal standard used here and asked whether it

matched the author's meaning; the author then asked for this deliverable, and the definition is recorded as declared (D1 in the register). Spelling in the quoted conjecture was corrected at the author's standing invitation.

Search before building. The corpus repository [9] was read before anything was written. Kernel K1 already proved the first cut; R8 already proved the collision, forced abstention and adequate-coordinate results; the frame kernel already defined residual selection; Before the Proof already proved bridge underdetermination. All four were reused unchanged. Axiom as Pharmakon already proved the open-versus-independent distinction for derivations, but it builds on Mathlib, so it is mapped by correspondence and not recompiled. The Lambda Dictionary [8] fixed the terms *access*, *will*, *stop* and *holder*, and it already held the accumulator's psychological and consciousness applications as bridges; the note uses those terms as fixed there. The corpus essay *Agency at the Frame Edge* [7] was read in full.

Errors found and fixed (recorded once).

1. *Deprecated names.* The toolchain marks `dif_pos`, `if_pos` and `if_neg` as deprecated, which the zero-warning rule rejects. They were replaced by `dite_eq_left`, `ite_eq_left` and `ite_eq_right`, with the same statements.
2. *An interface change.* `Option.noConfusion` now takes a heterogeneous equality, so two refutations of `none = some _` were rewritten as case analyses.
3. *A parse.* A tactic block inside an anonymous constructor absorbed the next component; parentheses fixed it.
4. *Classical choice in a witness.* A membership check decided by `decide` brought `Classical.choice` into one theorem of the worked model. The check was replaced by an explicit proof, and the audit script now rejects `Classical.choice` outside the four declarations of the classical hinge.

This document. `build_paper.py` fills every number, hash and kernel report in the text from the receipts, then runs three XeLaTeX passes. The build fails on any line more than 1 pt too wide, any missing glyph and any undefined reference. The LaTeX source, the build script and the kernel report are attached to this PDF.

References

[1] Lu Semita (EmergenceByDesign) et al. *Closure at the Adjudication Seam*, R8, complete proofs, 24 October 2026. Corpus repository [9], `WIP_docs/IRRProofEngine/DERIVATIVE/Closure_at_the_Adjudication_Seam_2026-10-24_R8_Complete_Proofs.zip`.

[2] Lu Semita (EmergenceByDesign). *Lambda, Reference, and the Spatiotemporal Descriptive Frame*, receipt edition V1, 10 September 2026. Corpus repository [9], `WIP_docs/IRRProofEngine/DERIVATIVE/Lambda_Reference_and_Spatiotemporal_Descriptive_Frame_V1.pdf`.

[3] Lu Semita (EmergenceByDesign), developed with Claude (Anthropic) within MetaLabe. *The Exchange Lock and Its Key*, laboratory note with Lean kernel K1, and its supplement, 31 October 2026. Corpus repository [9], `https://github.com/EmergenceByDesign/The-Lambda-Principle-of-Irreducibility/tree/main/WIP_docs/IRRProofEngine/DERIVATIVE/DIY_Quantum_Computing`.

[4] Lu Semita (EmergenceByDesign) and Orion. *The Double-Edged Receipt Standard: Quantifier Asymmetry, Scope, and Symmetric Burdens of Proof*, review draft v0.3, 23 September 2026. Corpus repository [9], `WIP_docs/Double_Edged_Receipt_Standard_v0_3_REVIEW_DRAFT-1.pdf`.

[5] Lu Semita (EmergenceByDesign) et al. *Axiom as Pharmakon: Scoped Deductive Closure, Symmetric Adjudication, and Class-Level Uncertainty*, edition R3 with Lean supplement, 30 October 2026. Corpus repository [9], `WIP_docs/IRRProofEngine/DERIVATIVE/`.

[6] Lu Semita (EmergenceByDesign). *Before the Proof*, Certified Edition V1, with Certified Appendix (bridge underdetermination), 21 August 2026. Corpus repository [9], `WIP_docs/IRRProofEngine/DERIVATIVE/Before_the_Proof_CERTIFIED_PACKAGE_2026-08-21/`.

[7] Lu Semita and associated working notes. *Agency at the Frame Edge: Will as Residue-Preserving Determination*. Corpus repository [9], `WIP_docs/Agency_at_the_Frame_Edge_Will_as_Residue_Preserving_Determination.md`.

[8] Lu Semita (EmergenceByDesign). *Lambda Dictionary*, R4. Corpus repository [9], `WIP_docs/IRRProofEngine/DERIVATIVE/Lambda_Dictionary_R4.md`.

[9] EmergenceByDesign. *The Lambda Principle of Irreducibility*, corpus repository. `https://github.com/EmergenceByDesign/The-Lambda-Principle-of-Irreducibility`.

[10] Lu Semita (EmergenceByDesign), developed with Claude (Anthropic) within MetaLabe. *Lean packet for kernel HP1*: kernel, reused modules, receipts, claim register and ledger, one-paste bundle, 31 October 2026. `HP1_Lean_Packet_Hard_Problem_Undecidable_2026-10-31.zip`, SHA-256 in Appendix A.

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